

IBM Cloud Infrastructure Blueprint — ROKS Cluster + VPC + Storage + Security



ROKS CLUSTER

Region: EU-DE Frankfurt (eu-de-1/2/3)

3 × bx2d-4x16 worker nodes

12 vCPU / 48 GB RAM / 900 GB SSD

13 namespaces (8 solutions + 5 shared)

50% savings promo applied at checkout

RH Partner Sub: eliminates RHEL cost



VPC & NETWORKING

1 VPC (i3-ea-vpc) in EU-DE

3 subnets (one per AZ)

Application Load Balancer for ingress

Private endpoints → no public egress cost

IBM Cloud Internet Services (CIS) CDN

Security Groups + Network ACLs (deny-by-default)



STORAGE STACK

IBM COS One-Rate: 75% discount perpetual ★

IBM Block Storage VPC: 500 GB (Maximo DB)

IBM File Storage VPC: 200 GB (shared)

MinIO (open source): 100 GB on worker SSD

IBM Container Registry: 5 GB free tier

IBM Backup VPC: automated Maximo backup



SECURITY ZONES

IBM Security Verify (Zero Trust IAM)

QRadar SIEM on Cloud (managed)

Red Hat Satellite → IEC 62443 compliance

Wazuh + Falco + OpenSCAP (open source)

IBM Guardium Data Protection (trial)

IBM Cloud Security Advisor (unified posture)

8 Agentic AI Solutions — Namespace-Separated on IBM Cloud ROKS

One shared cluster • Shared watsonx.ai inference • Shared Event Streams Kafka • Unified Grafana dashboard • LangGraph cross-agent orchestration

gridsentinel



GridSentinel AI

TANESCO (TZ)

LSTM + IEC 61850

<200ms fault detect

maxar



MaxAR Field Eng.

KenGen (KE)

CV + HoloLens 2

Auto Maximo WO <2s

aquasense



AquaSense AI

NCWSC (KE)

VAE Acoustic + IoT

87% leak detect

otdigitaltwin



OT Digital Twin

TANESCO+KenGen

Monte Carlo + watsonx

Asset failure predict

otshield



OTShield Security

All Clients

QRadar + Wazuh + SCAP

IEC 62443 auto-report

railmind



RailMind AI

KRC/SGR (KE)

LSTM + Watson IoT

Derailment prevention

airportos



AirportOS AI

KAJ/JKIA (KE)

NLP + Flight AI

Gate optimization

assetchain



AssetChain

All Clients

IBM Blockchain + Maximo

Immutable asset audit

Maximise Resources — Open Source + IBM Free Tiers + SaaS Trials

OPEN SOURCE STACK

- ArgoCD**
GitOps controller — auto-deploy all Helm charts from Git
- Grafana + Prometheus**
Unified monitoring + all 8 solution dashboards
- MinIO**
S3-compatible storage on ROKS worker SSD (no extra cost)
- LangGraph**
Multi-agent orchestration — <500ms cross-solution cascade
- Wazuh SIEM**
Host-based IDS + log aggregation → feeds QRadar
- Falco**
Runtime container anomaly detection (all namespaces)
- OpenSCAP**
IEC 62443 compliance scan → auto-generates PDF report
- MLflow**
ML experiment tracking for all 8 AI models
- Keycloak**
IAM fallback if IBM Verify quota exceeded
- Istio + Cert-Manager**
mTLS between namespaces + TLS auto-provisioning

IBM FREE TIERS (Sign Up Now)

- Watson Studio Lite**
20 CUH/mo + GPU — all synthetic data notebooks FREE
- IBM Event Streams Lite**
Managed Kafka — 75 GB storage, 5 partitions FREE
- Watson IoT Lite**
500 devices, 200 MB/day — all 8 device types FREE
- IBM AppID/Verify Lite**
1,000 users — all demo + event auth FREE
- watsonx.ai Lite**
20 CUH/month — prototyping + Prompt Lab FREE
- IBM Db2 Warehouse Lite**
1 GB analytics warehouse — Grafana data FREE
- IBM Container Registry**
5 GB free — all 8 solution Docker images FREE
- IBM Cloud Security Advisor**
Included with account — unified posture FREE
- IBM Cloud Shell**
Browser CLI with oc/kubectl — admin tasks FREE
- watsonx.data Lite**
Data lakehouse — query all synthetic datasets FREE

SAAS 30-DAY TRIALS (Activate Before Workshop)

- IBM Video Streaming**
Enterprise live streaming — 5,000 viewers FREE 30 days
- IBM Cloud Internet Services**
CDN + DDoS for event portal FREE 30 days
- IBM Cognos Analytics**
Executive dashboards for client LOI packages FREE
- IBM OpenPages GRC**
IEC 62443 compliance framework integration FREE
- IBM Blockchain Platform**
Hyperledger Fabric for AssetChain FREE trial
- IBM Guardium**
Data security for Maximo DB FREE 30 days
- watsonx.governance**
AI model factsheets + bias detection FREE trial
- StreamYard Basic**
Multi-speaker broadcast studio FREE (2 speakers)
- Airmeet Free**
Virtual networking rooms FREE up to 100 attendees
- Slido Free**
Live polls + Q&A FREE up to 200 participants

IBM Video Cloud + Hybrid Event Platform — IBM Think/TechXchange Model

March 30–31 2026 • Live Streaming • On-Demand Replays • Audience Engagement • Lead Generation from 83 East Africa Orgs



StreamYard (Free)

Web-based multi-speaker studio — remote IBM speakers join via browser. No software install needed.

Slido (Free 200 pax)

Live polls + Q&A for all 11 sessions — unified for in-room + virtual attendees simultaneously.

Airmeet (Free)

Virtual networking rooms, IBM/RH sponsor booths, 1:1 networking for up to 100 virtual attendees.

Mailchimp (Free)

Post-event: auto-email replay links to all 83 registered organizations. FREE up to 2,000 contacts.

IBM AppID Gated Access

Email-gated on-demand replays — every replay view from 83 orgs is a tracked qualified prospect.

GitHub Pages Portal

Post-event replay portal at i3tech.co.ke — embeds IBM Video player. FREE hosting via GitHub Pages.

Red Hat Owned Assets + LangGraph Innovation — Untapped Value You Already Own

REDHAT SUBSCRIPTIONS — FULLY OWNED (Oct 2026 / Mar 2027)

RH Partner Sub (MW02049)

500 Nodes • Expires: Oct 2026 • ~\$50K+ value

Eliminates ALL ROKS worker RHEL licensing. Attach at Hour 1 provisioning.

RH Satellite (MCT3718)

50 Nodes • Expires: Oct 2026 • ~\$5K+ value

Patch management → IEC 62443 compliance auto-report → competitive differentiator.

RH Beta Access (RH00069)

All team • Expires: Mar 2027 • Strategic

Early OCP/RHEL AI features — potential next-gen watsonx operator previews.

RH Developer Subs x15 (RH00798)

15 devs • Expires: Sep 2026+ • ~\$1.5K value

Full RHEL entitlements for all team — distribute NOW for local dev environments.

RH Partner Support (MW02040)

5 contacts • Expires: Oct 2026 • ~\$2.5K value

Direct RH escalation during 48-hr sprint — register contacts before starting.



LangGraph MULTI-AGENT INNOVATION

One fault event → 3 intelligent agents respond simultaneously in <500ms

1 GridSentinel detects TANESCO grid fault

2 LangGraph event routed to 3 parallel agents

3 OT Digital Twin → Monte Carlo risk assessment

4 OTShield → auto-generates IEC 62443 report

5 MaxAR → dispatch work order to field engineer

REGULATORY DIFFERENTIATOR: No competitor in East Africa has automated IEC 62443 compliance reporting. Satellite → Wazuh → QRadar → OTShield → PDF in <5 seconds.

Credit Budget Optimisation — \$8,000 IBM Cloud Credits | Maximum Value Strategy

ROKS Cluster (3 nodes, 50% promo)

Est. \$200–300 / month

\$200–300

watsonx.ai Runtime (Standard)

Est. \$300–500 / month

\$300–500

IBM API Connect (ROKS-deployed)

Est. \$80–120 / month

\$80–120

QRadar SIEM on Cloud (lite)

Est. \$100–150 / month

\$100–150

IBM COS One-Rate (75% perpetual promo)

Est. <\$20 / month

<\$20

Maximo MAS 9.1 (AppPoints Essentials)

Est. \$400–500 / month

\$400–500

IBM Event Streams (upgrade from Lite)

Est. \$150–250 / month

\$150–250

VPC + Networking + Storage (Block/File)

Est. \$150–200 / month

\$150–200

Total Sprint
Spend (60 days)

\$1,800–2,400

Well within \$8K budget

FREE / Trial
Components Value

\$60,000+

Open source + RH subs + IBM free tiers

Credits Remaining
Post-Sprint

\$5,600–6,200

Available for first client pilot

First Client Pilot
Enabled

TANESCO or KenGen

Paid pilot from remaining credits

48-Hour Production Sprint — Hour-by-Hour Implementation Plan

PHASE 1 H0-6 Foundation	TRACK A H6-18 AI Services	TRACK B H6-18 Infra & Security	PHASE 3 H18-36 Integration	PHASE 4 H36-48 Demo Hardening
Philip + DevOps	AI Lead	Infra Lead	All Team	All Team
Log in, set \$2,500 budget alert	Deploy watsonx.ai on ROKS	Deploy Maximo MAS 9.1	GridSentinel: LSTM pipeline	IBM Video Streaming trial
Sign up COS One-Rate ★	Create 8 watsonx.ai projects	Load KenGen asset data CSV	MaxAR: CV → Maximo WO	OBS Studio 3-scene setup
Provision ROKS + 50% promo	GridSentinel synthetic data H9	Red Hat Satellite setup	AquaSense: VAE → Grafana	Slido polls for all sessions
Attach RH Partner Sub (MW02049)	AquaSense VAE data H10	Deploy Grafana + Prometheus	OT Twin + OTShield wiring	Load test 100 concurrent req.
Create 13 namespaces	Deploy API Connect + ApplID	QRadar + Wazuh + Falco	RailMind + AirportOS + AssetChain	6 client demo scripts ready
Deploy ArgoCD + Kafka topics	LangGraph multi-agent setup	IEC 62443 OpenSCAP setup	7 integration tests (H28-H35)	3-min fallback video recorded